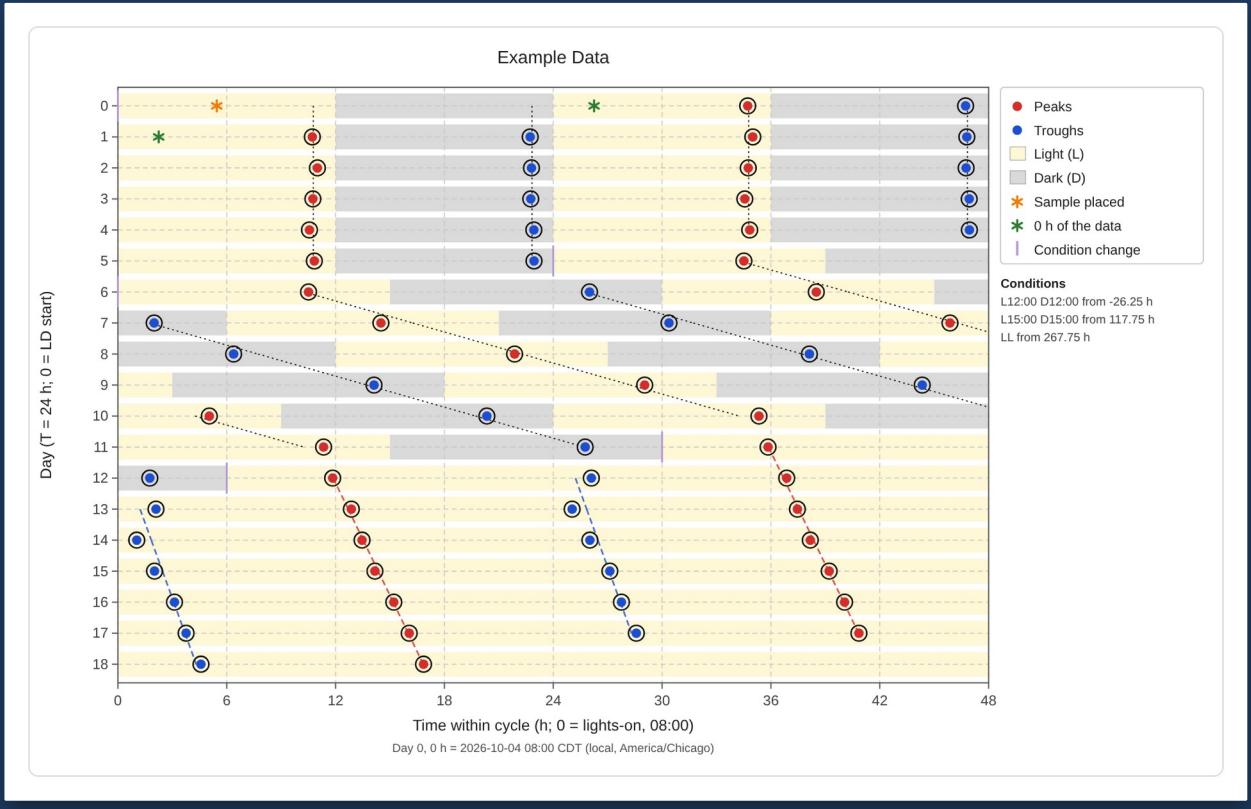
Actogram Plotter

User Manual · Web version

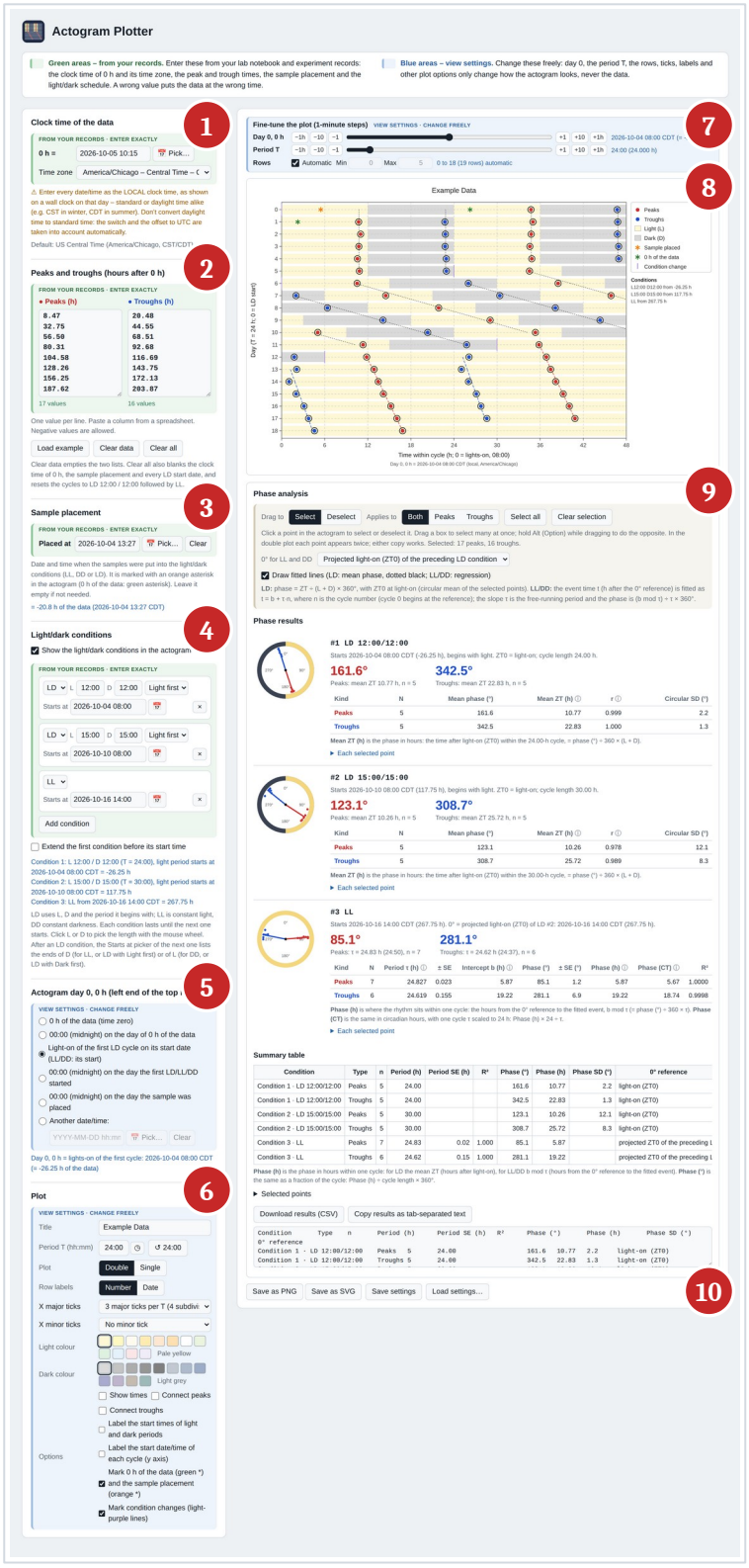
Draws an actogram from the peak and trough times of a circadian time series, such as a bioluminescence rhythm recorded with JKtron or Ttron – typed in, or detected in the time series itself – together with the light/dark schedule of the experiment.

Open the app: https://dxda6216.github.io/actogram_plotter/



Runs in any current desktop browser · No installation · Your data stays in your browser

Screen overview



1 Clock time of the data
date and time of 0 h, time zone

2 Peaks and troughs
paste the times, or detect them in a time series

3 Sample placement
when the samples went into the conditions

4 Light/dark conditions
one card per LD, LL or DD condition

5 Actogram day 0, 0 h
left end of the top row

6 Plot
period T, ticks, colours, labels, options

7 Fine-tune the plot
day 0, period T and rows in 1-minute steps

8 Actogram
the figure, updated as you type

9 Phase analysis
select points; period and phase results

10 Save buttons
PNG, SVG and settings files

With time-series data, a time-series plot also appears above the fine-tune panel (page 11).

Green and blue areas



The settings column and the fine-tune panel are tinted in two colours. They tell you which values must come from your experiment records and which you can change as you like.

Green – from your records

Enter exactly, from the lab notebook

- Clock time of the data: 0 h and the time zone
- Peaks and troughs, or the time series
- Sample placement: placed at
- Light/dark conditions: one card each with the condition, L, D, light or dark first and the start date and time

A wrong value puts the data, the lighting or the asterisks at the wrong time. Check these against your records before you save a figure.

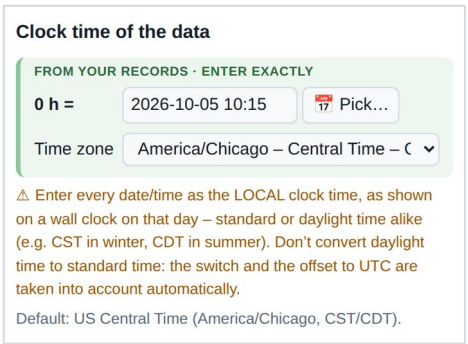
Blue – view settings

Change freely

- Actogram day 0, 0 h
- Plot: title, period T, double or single plot, row labels, ticks, band colours and options
- Fine-tune panel: day 0, period T and rows

These only change how the actogram looks; the data and the schedule stay the same. Try different settings as often as you like.

0 h of the data is a recorded value, so it is set only in the left column. **Sand-coloured:** the selection tools and the 0° reference of Phase analysis. They change the analysis, not the data; the results below them are on white.



A green area: Clock time of the data



A blue area: the fine-tune panel

» Quick start

Before you start: open the app. It starts with example data (title “Example Data”). Click **Clear all** to start a new experiment. Steps 1–4 (green) come from your records; 5–7 (blue) only change the view; 9 analyses the points.

1 **Set the clock time of 0 h**
Pick the local date and time that corresponds to 0.00 h of your data. Keep the time zone at US Central Time unless the experiment ran elsewhere.

2 **Paste peaks and troughs, or a time series**
Paste the peak and trough times (hours after 0 h) into the two lists, or choose Detect in time-series data and paste the time series instead (pages 8–11).

3 **Enter the sample placement**
Pick when the samples were put into the light/dark conditions (optional; shown as an orange asterisk).

4 **Enter the light/dark schedule**
Describe the first condition (LD, LL or DD) and up to five changes, each with its start date and time.

5 **Choose actogram day 0, 0 h**
Usually “lights-on of the first LD cycle”, so the horizontal axis counts hours from lights-on.

6 **Set the plot**
Set period T (24:00 for a 24-h day) and the display options: ticks, labels, title.

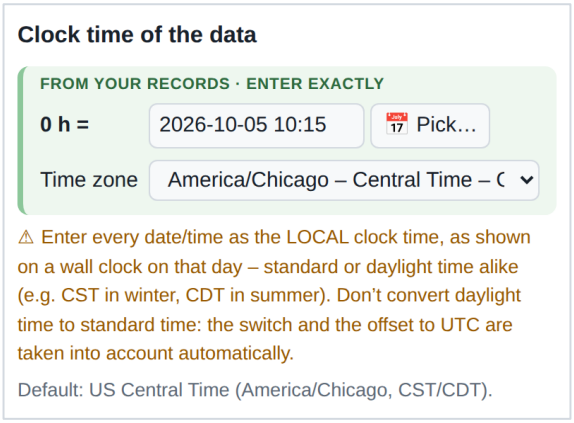
7 **Fine-tune if needed**
Above the actogram: shift day 0 or T in 1-minute steps, or choose which rows (cycles) are shown.

8 **Check the actogram**
The figure redraws after every change. Red points are peaks, blue troughs; asterisks mark 0 h and the placement, purple lines the condition changes.

9 **Analyse phases**
Under the actogram: select points and read the phase (and, in LL/DD, the period) of each condition.

10 **Save**
Save the figure as PNG or SVG, and your inputs as a settings file to reload later.

1 Clock time of the data



0 h = is the local date and time that corresponds to 0.00 h of your time-series data. Every peak and trough time is counted from this moment.

- **Data that start at 0.00 h** (the usual JKtron/Ttron output): enter the date and time when the measurement began.
- **Data whose first point is not 0.00 h:** subtract the time of the first data point from the date and time it was recorded.

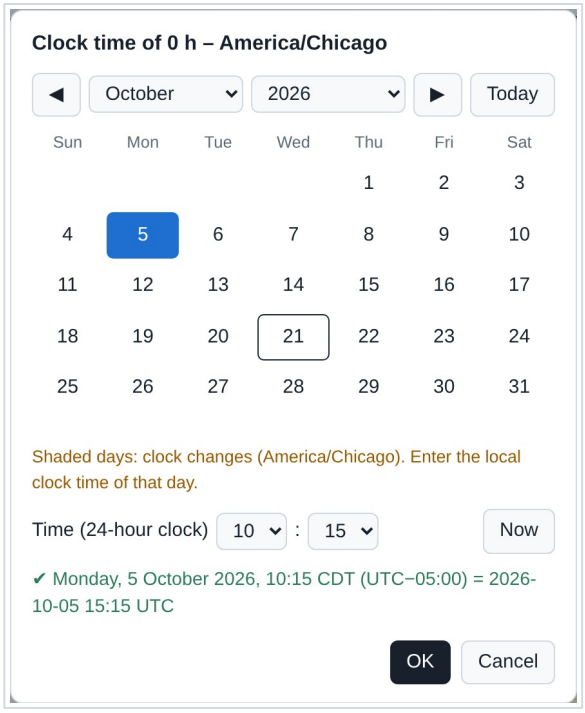
Example. The measurement began on 18 October 2026 at 13:32 (1:32 PM), and the first data point is labelled 3.00 h. Then 0 h = 13:32 − 3:00 = 10:32: click **Pick...** and select **2026-10-18 10:32**.



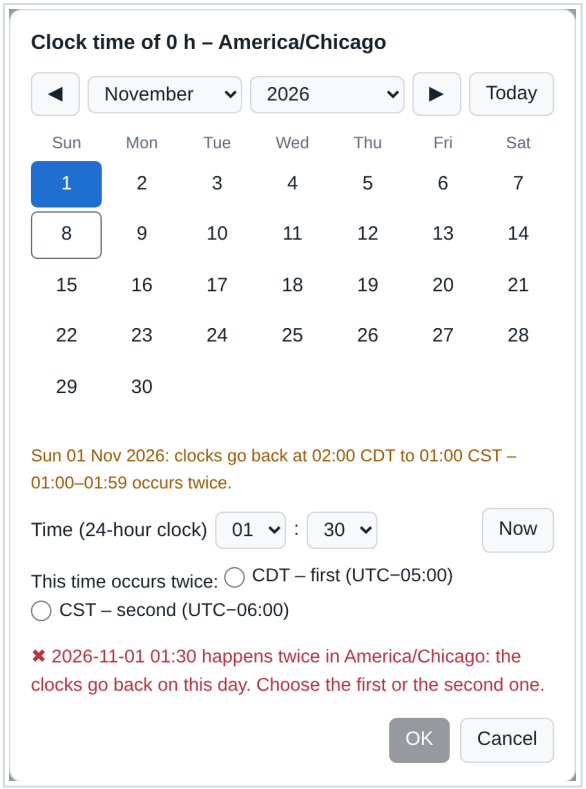
Time zone. Normally leave it at the default, US Central Time (America/Chicago). Change it only if the clock of the experiment ran in another zone; the list runs from east to west, with UTC last, and "Other…" accepts any zone name or an offset such as UTC+09:00.

Enter local clock times. Type every date and time as the wall clock showed it that day, CST in winter or CDT in summer. Do not convert daylight time: the app converts to UTC itself, so intervals across a daylight-saving change get their true length.

1 Using the date and time picker



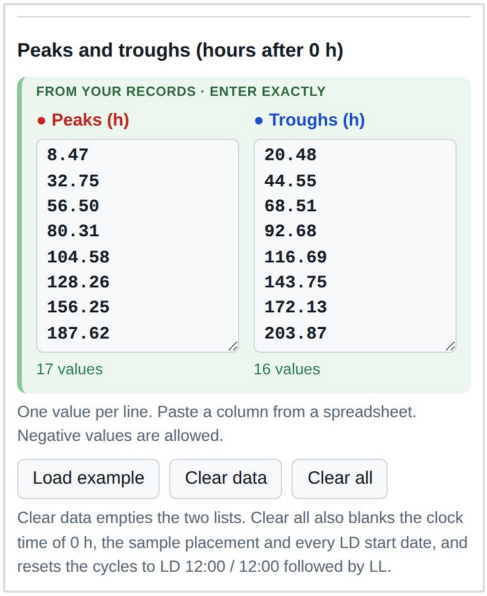
An ordinary day



The day the clocks go back (1 Nov 2026)

- **Open:** click the date field or **Pick...** (📅 in the LD table).
- **Date:** choose the month and year from the lists or with the arrow buttons, then click the day. **Today** jumps to today.
- **Time:** choose the hour and minute (24-hour clock); the mouse wheel over a list also changes it. **Now** sets the current date and time.
- **Check:** the green line shows the weekday, the local time with its zone abbreviation, the offset and the time in UTC. **OK** works only when the line is green.
- **Shaded days** are days on which the clocks change. A time in the skipped hour (March) is refused. For a time in the repeated hour (November, 01:00–01:59) choose the first (CDT) or the second (CST) occurrence; it is saved as, for example, “2026-11-01 01:30 CST”.

2 Peaks and troughs

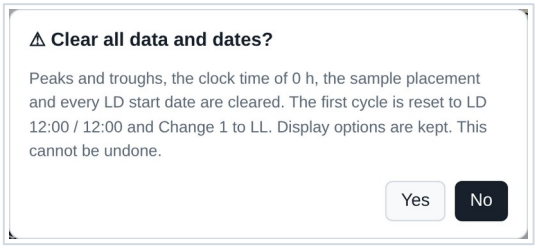


- Paste the **peak times** into the left list and the **trough times** into the right list, one value per line, in hours after 0 h (for example 12.47).
- Copy a column from Excel or from the output of your rhythm-analysis software and paste it with Ctrl+V. Commas, tabs and spaces also separate values. Negative values are plotted before 0 h.
- Under each list the app shows how many values it read (green) or which items are not numbers (red). Remove units or text, and use a decimal point.
- The two lists are independent: they may hold different numbers of values, in any order.
- Here the app does no rhythm analysis; it plots the times you enter. To detect them in a time series instead, see pages 8–11.

Load example replaces the data, the dates and the LD cycles with the example data set. Display options are kept.

Clear data empties both lists (Ctrl+Z in a list brings them back).

Clear all also blanks the clock time of 0 h, the sample placement and every LD start date, and resets the cycles to LD 12:00 / 12:00 followed by LL: the starting point for a new experiment.

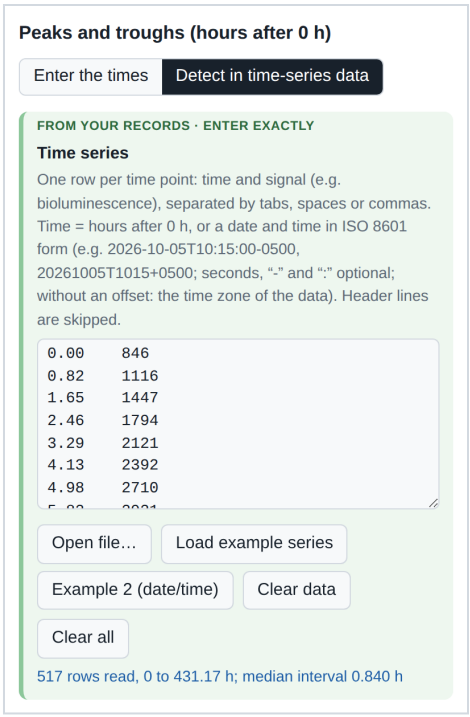


Each button asks first; No is preselected

Two ways to give the times. The switch at the top of this block, **Enter the times** | **Detect in time-series data**, chooses between typing the two lists (this page) and detecting the peaks and troughs in a time series (pages 8–11; the default). With time-series data, Clear data and Clear all also empty the time series.

2 Peaks and troughs from a time series

Instead of typing the peak and trough times, you can give the app the time series itself and let it find them. **Detect in time-series data**, at the top of the **Peaks and troughs** block, is the default: the app starts with the example time series. **Enter the times** brings back the two lists of the previous page.

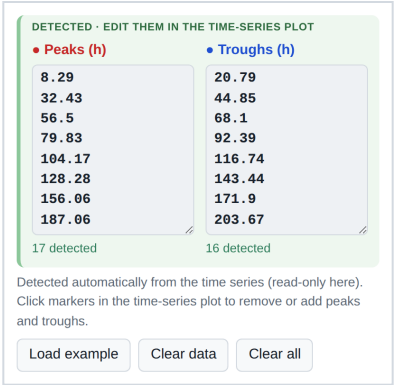


The input switch and the time series

- **Format:** one row per time point. Column 1 = time in **hours after 0 h** of the data (Clock time of the data, page 5), or a **date and time** (next page); column 2 = the signal, e.g. bioluminescence counts. Tabs, spaces, commas or semicolons separate the columns; further columns are ignored.
- **Paste** two columns from Excel or a text file, **Open file…** to read a .txt, .csv, .tsv or .dat file, or **Load example series** for a 431-h bioluminescence record recorded under the example schedule; it also loads the example clock time, sample placement and LD cycles (LD 12:12 → LD 15:15 → LL).
- **Header and note lines** (text, or lines starting with #) are skipped. NaN, NA or – in the signal column is a missing value.
- The **blue lines** under the box report how many rows were read, the time span and the interval; **red lines** report problems.

Unequal intervals. If the intervals differ by more than ±5 % from the median interval, the data are first resampled by linear interpolation onto a regular grid with the median interval, gaps longer than 2.5 intervals staying empty, and the detected times are grid times. The intervals of the example (0.80–0.87 h, median 0.84 h) are within ±5 %, so it is used as it is.

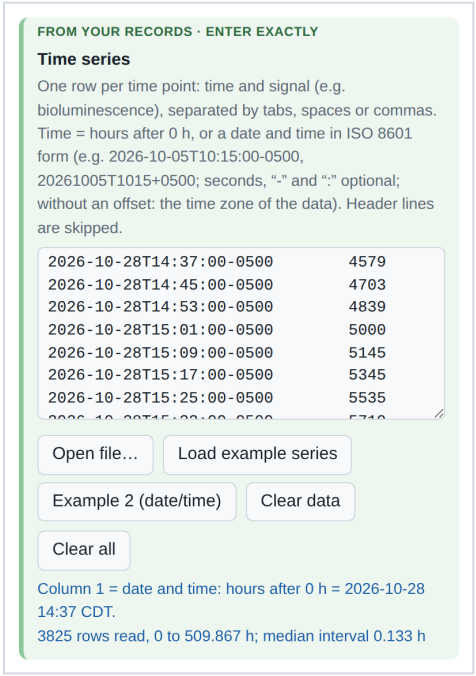
- The **Peaks** and **Troughs** lists now show the detected times (tagged "Detected", read-only). They feed the actogram and Phase analysis exactly like typed values. Under each list: how many were detected and how many were added by hand.
- Newly detected points are selected for Phase analysis automatically; a point you deselect stays deselected.
- **Clear data** and **Clear all** are also under the time series: Clear data empties the time series and its hand edits; Clear all also blanks the dates and resets the LD cycles. **Enter the times** brings back the lists you typed earlier in this session.



The detected lists

2 Date and time in column 1

Column 1 may also hold a **date and time** (ISO 8601) instead of hours. The app turns each one into hours after the **Clock time of 0 h** (page 5), so the data, the peaks and troughs and the light/dark schedule line up by themselves.



Example 2 (date/time)

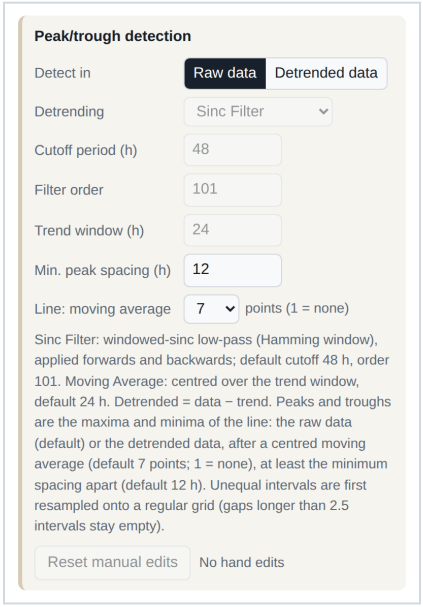
Form	Example
With - and :, seconds	2026-10-05T10:15:00-0500
Without seconds	2026-10-05T10:15-0500
Without - and :	20261005T101500-0500
Without seconds	20261005T1015-0500
Offset from UTC	-0500, -05:00, -05, +0900, Z (= UTC)
Space instead of T	2026-10-05 10:15:00
Fraction of a second	2026-10-05T10:15:00.5-0500
No offset	2026-10-05T10:15: local clock time of the **Time zone** of the data

- **The offset** is the difference from UTC: −0500 = 5 h behind UTC (e.g. CDT), +0900 = 9 h ahead (e.g. JST). With an offset every time is exact, whatever the Time zone setting; a daylight-saving change during the record is no problem.
- **0 h:** the hours count from the Clock time of 0 h. If it is not set, the first time point is used as 0 h and the blue line says so. To start the data at 0.00 h, set the clock time of 0 h to the first time point.
- **Mixed rows:** if most rows have a date and time, rows with plain hours are skipped (and the other way round).
- **Example 2 (date/time)** loads a record taken **every 8 minutes** from 2026-10-28 14:37 (3825 rows) with its own schedule: LD 12:12 for 5 days, LL 5.5 days, LD 11.5:11.5 (T = 23 h, dark first) 6 days, DD 5 days (free-running period about 25.05 h). The offset changes from −0500 to −0600 when daylight saving time ends (1 Nov). It sets the Filter order to 301: the order counts samples, so short intervals need a higher order.

Check the blue line. "Column 1 = date and time: hours after 0 h = 2026-10-28 14:37 CDT" shows which moment is 0 h, and the next line the hours covered (here 0 to 509.87 h, every 0.133 h). If the hours look wrong, check the clock time of 0 h, the offsets and the Time zone.

2 Detecting peaks and troughs

The sand-coloured box under the time series sets how the peaks and troughs are found. Like the selection tools of Phase analysis, it changes the analysis, not the data: try different settings as often as you like.



Peak/trough detection

Setting	Default	What it does
Detect in	Raw data	Find the peaks in the **raw data**, or in the **detrended data** (data − trend).
Detrending	Sinc Filter	How the trend is found: Sinc Filter or Moving Average.
Cutoff period (h)	48	Sinc Filter: changes slower than this period count as trend.
Filter order	101	Sinc Filter: filter length in samples; raise it for short intervals (301 for 8 min).
Trend window (h)	24	Moving Average: width of the centred window used as trend.
Min. peak spacing (h)	12	Two peaks (or two troughs) must be at least this far apart.
Line: moving average	7 points	Centred moving average of the line the peaks are found on: 1 (none), 3, 5 … 15 points.
Reset manual edits	–	Discards the hand edits (asks first).

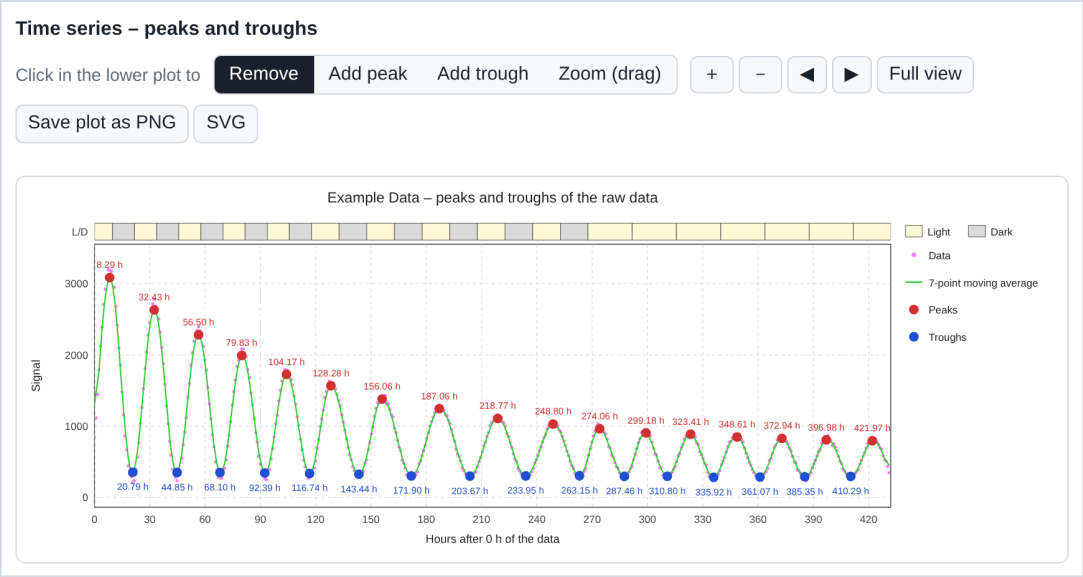
How the detection works

1. **Prepare the data.**
   Rows are sorted by time, duplicate times averaged, and unequal intervals resampled onto a regular grid (page 8).
2. **Remove the trend (detrended data only).**
   Sinc Filter: a windowed-sinc low-pass filter (Hamming window), applied forwards and backwards so the curve is not shifted in time. Moving Average: a centred mean over the trend window. Detrended = data − trend.
3. **Draw the line.**
   A centred moving average over the chosen number of points (default 7; 1 = the data themselves).
4. **Find the extremes.**
   Every local maximum of the line is a peak candidate and every local minimum a trough candidate. Of two candidates closer than the minimum spacing, the higher peak (lower trough) is kept.
5. **Apply your hand edits.**
   Removed points are taken away and added points put in (next page).

Tip. Too many peaks from small wiggles? Use a longer moving average or a larger minimum spacing, or remove them by hand. With a strongly falling baseline (damping), detrended data often give more regular peaks.

2 The time-series plot

With **Detect in time-series data** chosen, a plot of the time series appears above the fine-tune panel. It shows what was detected and lets you correct it by clicking.

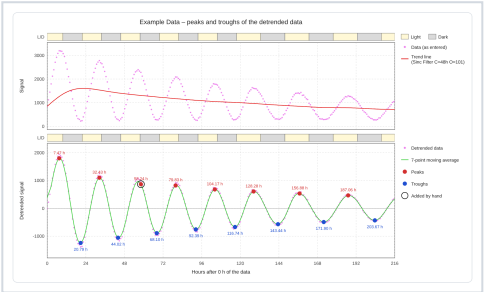


Raw data (default): the line is a 7-point moving average; bars above: light and dark

- **Lower plot:** the data used for detection (violet points), the line (green) and the peaks (red) and troughs (blue) with their times. With detrended data an **upper plot** also shows the data as entered with the trend line (red).
- **L/D bars:** when **Show the light/dark conditions** is ticked and the schedule is valid, a bar above each plot shows the light and dark periods in the Light and Dark colours of the Plot settings.

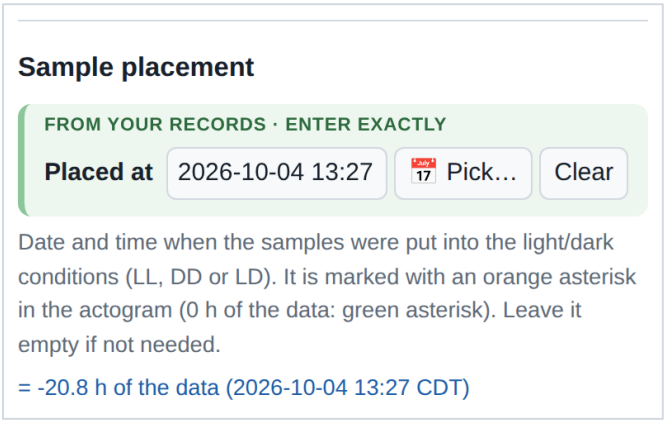
Control	What it does
Remove	Click near a marker (within about 25 pixels) to delete it.
Add peak / Add trough	Click near the line: the marker snaps to the nearest sample. A point that is already a peak or trough is refused.
Zoom (drag)	Drag across a time range to show only that range.
+ / −, arrows, Full view	Zoom in or out, scroll back or forward in time, show the whole record.
Save plot as PNG / SVG	Saves the time-series plot as a picture.

- **Hand edits** are marked with a black ring. They are remembered as times and kept when you change the data or the detection settings: a removed point stays removed if it is found again; an added point snaps to the new line.
- **Reset manual edits** returns to the automatic detection. The settings file saves the edits too.
- **Desktop version: Show plot** opens the plot in its own window with the zoom/pan toolbar; switch zoom and pan off before clicking markers.



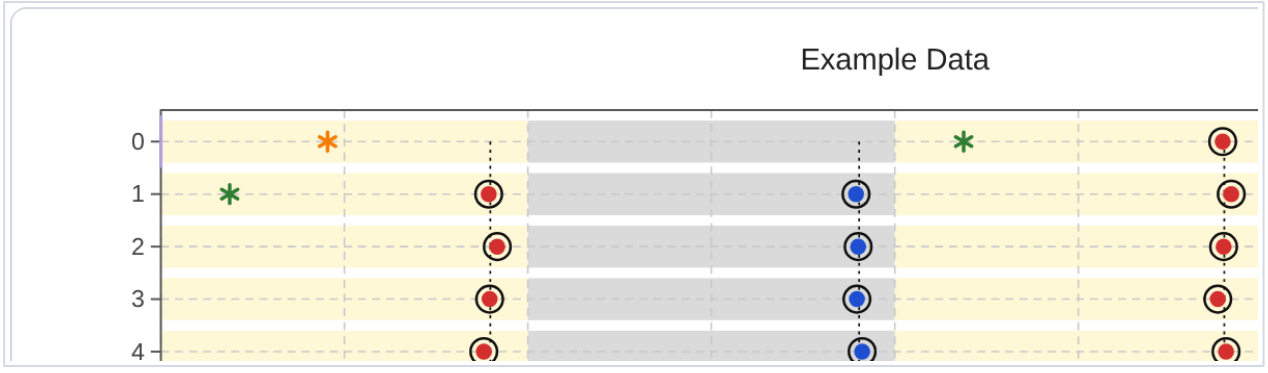
Detrended, zoomed: peak at 56.50 h removed, one added at 58.24 h

3 Sample placement



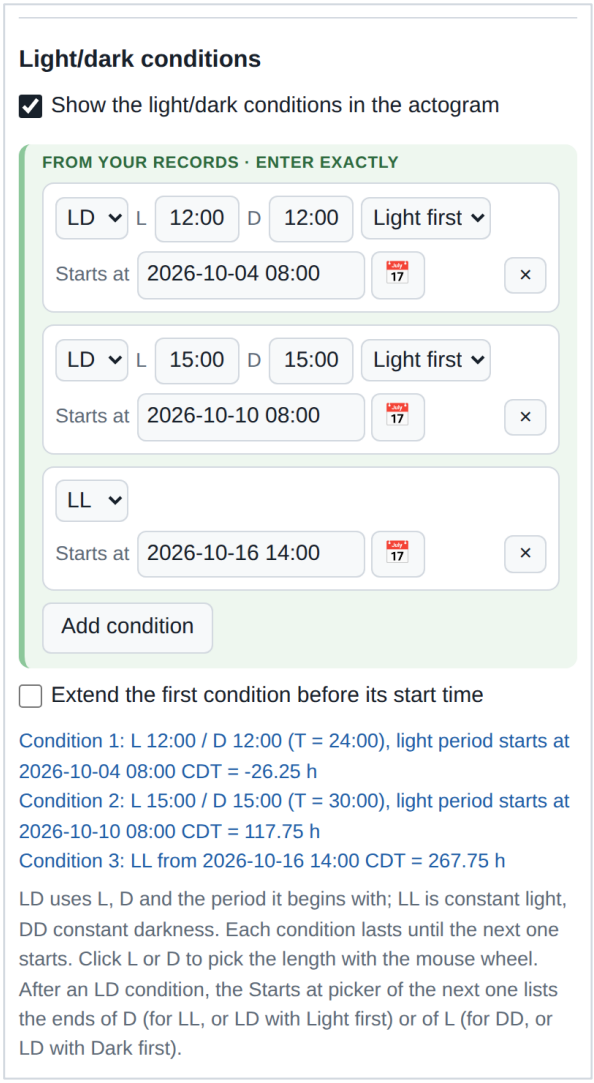
Placed at is the local date and time when the samples were put into the light/dark conditions (LL, DD or LD), for example when the culture dishes were set in the incubator. It is optional: leave it empty if you do not need it, and use **Clear** to empty it.

- The blue line under the field shows the same moment in hours of the data (for the example, −20.8 h: 20.8 hours before 0 h).
- In the actogram it is marked with an **orange asterisk**; 0 h of the data is marked with a **green asterisk**. In a double plot each asterisk also appears one row higher, at x + T.
- With Rows set to Automatic, the row that contains the placement is always shown.
- The placement also sets one of the choices for **Actogram day 0, 0 h**: 00:00 (midnight) on the day the sample was placed.
- Untick **Mark 0 h of the data … and the sample placement** under Plot to hide both asterisks, for example in a figure for publication.



*Example data: orange * = samples placed (4 Oct 13:27), green * = 0 h of the data (5 Oct 10:15)*

4 Light/dark conditions

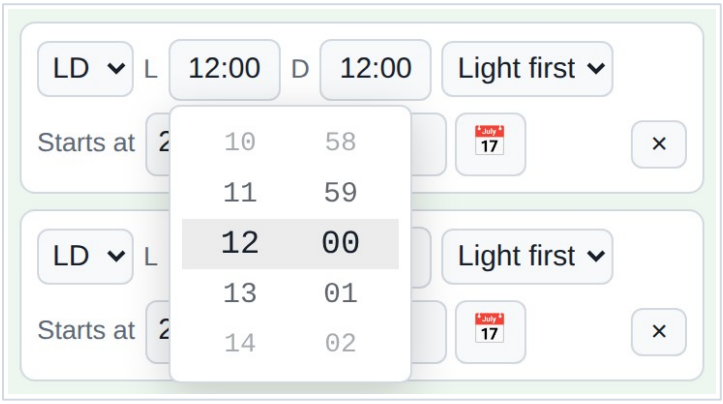


Each condition has its own card. Enter them from your records, in time order.

Item	How to set it
Condition	LD = light/dark cycle, LL = constant light, DD = constant darkness. LL and DD hide L and D.
L, D	Lengths of the light and dark periods, 00:00–99:59 (default 12:00); 15:00 + 15:00 gives T = 30 h. Click the field to pick with the mouse wheel, or type.
Light first / Dark first	Whether an LD cycle starts with its light or its dark period at its start time.
Starts at	Local date and time at which the condition begins (📅 picker).
×	Removes the condition; the later ones move up.
Add condition	Adds a card at the end (up to six conditions). Each lasts until the next one starts.
Show the light/dark conditions	Draws light (pale yellow) and dark (light grey) bands behind the points.
Extend the first condition before its start time	Draws the first condition backwards, e.g. entrainment before recording.

Blue lines under the cards show when each condition starts, in hours of the data; red lines report errors.

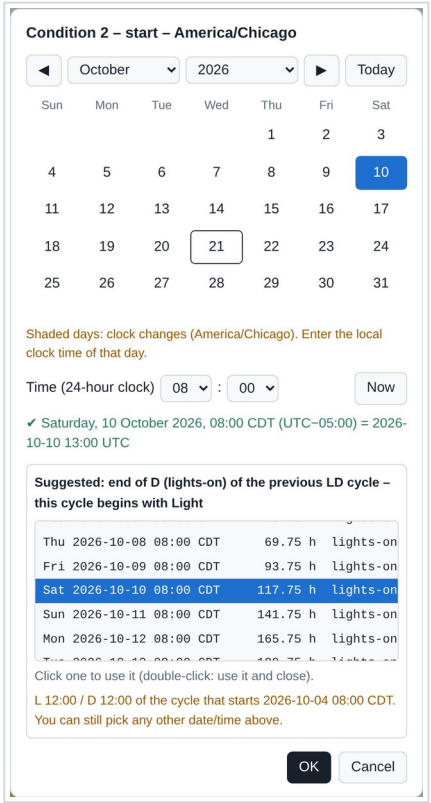
4 Lengths and condition changes



Picking L and D with the mouse. Click the L or D field. Scroll the mouse wheel over the hours or the minutes column, drag a column up or down, or click a number; the value in the grey band is used and the plot follows at once. Arrow keys also work. Press Esc or Enter, or click elsewhere, to close.

Suggested start times. When the condition before a change is LD, the start picker of that change also lists when that LD cycle's D or L ends. Click one to use it, or double-click to use it and close; any other time can still be picked. Set Condition and Begins with first.

New condition	Suggested times
LL	End of D (lights-on)
DD	End of L (lights-off)
LD, Light first	End of D (lights-on)
LD, Dark first	End of L (lights-off)

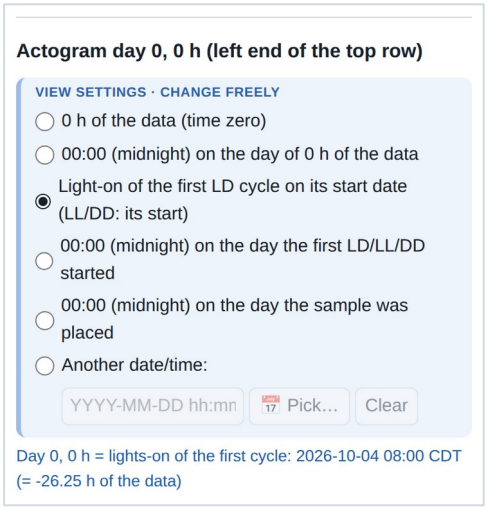


Change 1 (LD beginning with Light)

The schedule of the example data

	Condition	L / D	Starts at
First	LD	12:00 / 12:00	2026-10-04 08:00
Change 1	LD	15:00 / 15:00	2026-10-10 08:00
Change 2	LL	–	2026-10-16 14:00

5 Actogram day 0, 0 h



This sets the moment at the **left end of the top row** of the actogram. It is 0 h of the “Time within cycle” axis, and row n starts n × T later. It is a view setting: change it freely.

a **0 h of the data (time zero)**
Rows start at the clock time entered as 0 h.

b **00:00 (midnight) on the day of 0 h of the data**
Rows start at midnight of that day; the axis shows the time of day.

c **Light-on of the first LD cycle on its start date (LL/DD: its start)**
Rows start at lights-on, so the axis counts hours after lights-on (zeitgeber time). Recommended for most experiments.

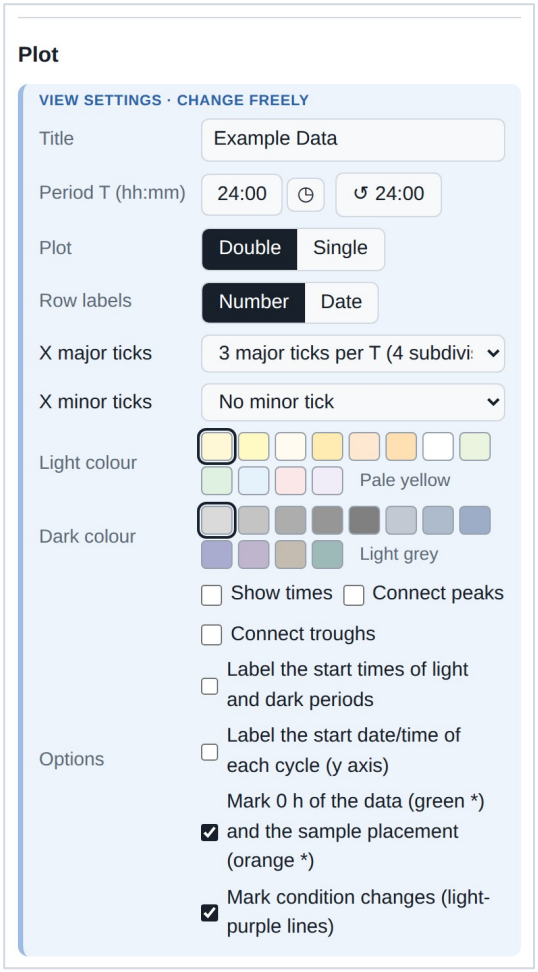
d **00:00 (midnight) on the day the first LD/LL/DD started**
Rows start at midnight of the day the lighting schedule began.

e **00:00 (midnight) on the day the sample was placed**
Rows start at midnight of the placement day (needs a sample placement time).

f **Another date/time**
Rows start at any date and time you pick. Clear empties the field.

Good to know. The blue line under the options shows the chosen moment in hours of the data. If the chosen reference is missing (LD cycles hidden or with an error, no first start date, or no sample placement time), 0 h of the data is used and a note says so. Shifting day 0 in the fine-tune panel switches this to “Another date/time”.

6 Plot settings

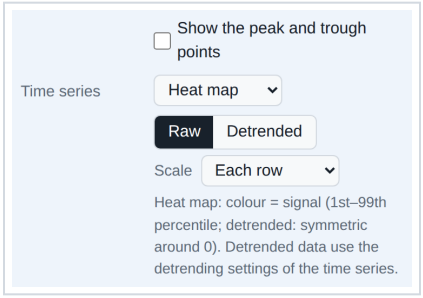


Setting	What it does
Title	Title above the figure (empty: a description of the plot).
Period T	Length of one row, 0:01–240:00 (default 24:00); ◷ wheel, ↺ 24:00 resets it. A double plot spans 0–2T.
Plot	Double (two cycles per row) or single.
Row labels	Row numbers or the date each row starts.
X major / minor ticks	0–5 labelled ticks per T (default 3) and 0–11 unlabelled ticks between them.
Light / Dark colour	12 swatches each (default pale yellow and light grey).
Show times	Writes the time (h) next to each point.
Connect peaks / troughs	Joins each point to the one about one period later.
Label the start times of light and dark periods	Local time and date of each lights-on and lights-off, e.g. “D 04:00 Oct. 6”.
Label the start date/time of each cycle	Each row’s start on the vertical axis.
Mark 0 h … and the sample placement	Green and orange asterisks (on by default).
Mark condition changes	Thin light-purple line where each LD, LL or DD condition starts (on by default).

Time-series data. Show the peak and trough points and the **Time series** options (activity bars or a heat map of the raw or detrended data in every row) are described on the next page.

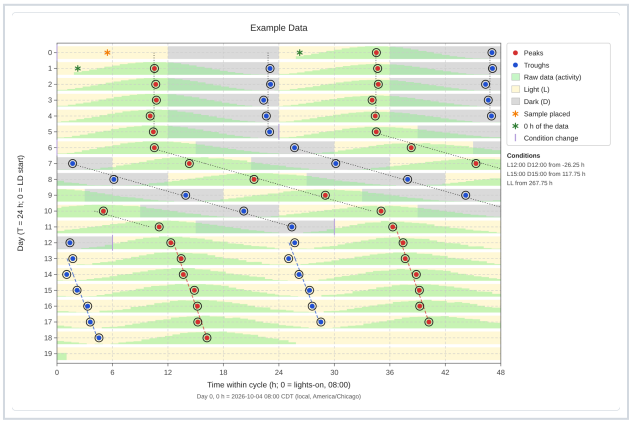
6 Drawing the time series in the actogram

With **Detect in time-series data** chosen, the actogram can also show the data themselves in every row, like a locomotor-activity record or as a heat map, behind the peaks and troughs. These are view settings under **Plot**.

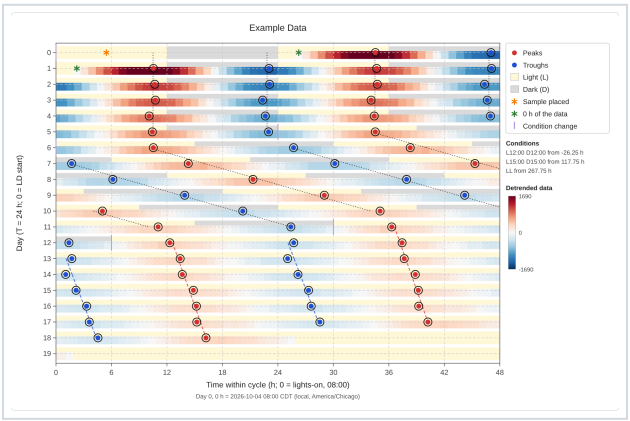


Plot: the new options

Setting	What it does
Show the peak and trough points	Untick to show only the data (on by default; also works with typed times).
Time series: None / Activity bars / Heat map	**None** (default). **Activity bars:** bar height = signal, like wheel-running activity, in light green (50 % transparent). **Heat map:** colour = signal (raw: dark blue → yellow; detrended: blue → white → red, white = 0), in the lower part of each row so the light/dark band stays visible above.
Raw / Detrended	The data drawn. Detrended data use the detrending settings of the time series (page 10), even when the peaks are found in the raw data.
Scale: Whole record / Each row	**Whole record:** one scale for all rows (1st–99th percentile), so damping is visible. **Each row:** every row is scaled on its own, so the waveform stays visible while the rhythm damps.



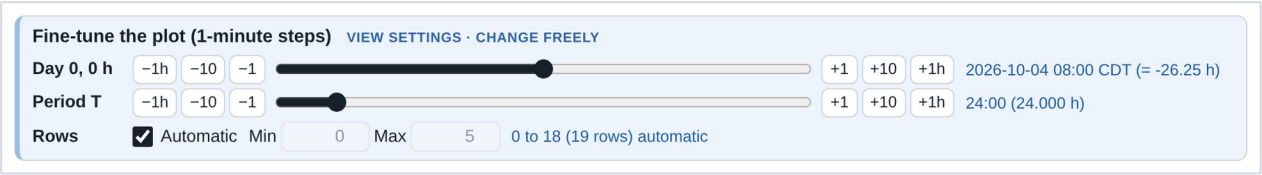
Activity bars, raw data, scaled per row



Heat map, detrended data, whole record

Good to know. In a double plot every sample appears twice, like the points. With Rows set to Automatic, the rows cover the whole time series. The heat map has a colour bar under the legend; Save as PNG / SVG include the data. Peaks, troughs, fitted lines and asterisks are drawn on top.

7 Fine-tuning the plot

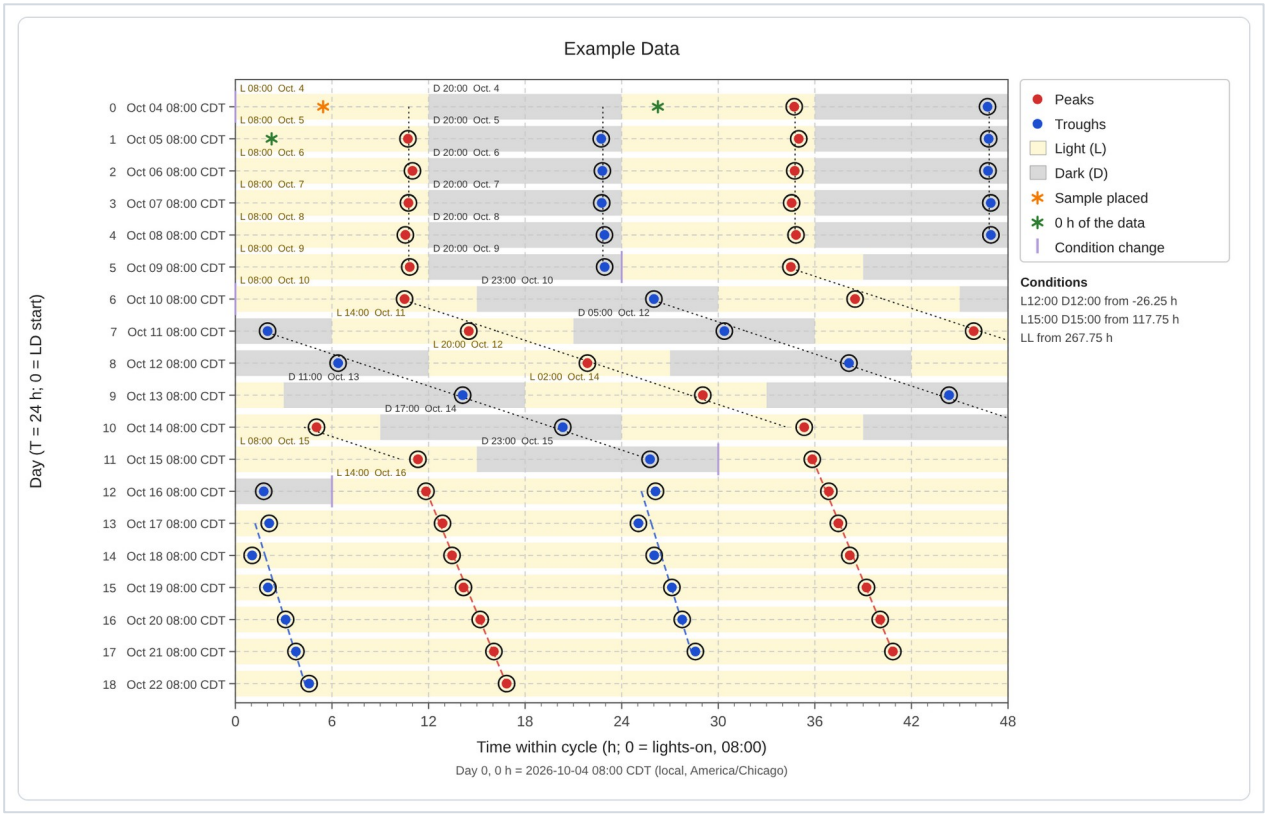


The panel sits above the actogram. Its controls change the view only.

Control	What it does
0 h of the data	Not in this panel: it is a recorded value, set only in the left column (Clock time of the data).
Actogram day 0, 0 h	Moves the left end of the rows. Using it switches day 0 to “Another date/time”.
Period T	Changes T between 0:01 and 240:00.
−1h −10 −1 / +1 +10 +1h	Change the value by 1 h, 10 min or 1 min. Hold a button to repeat.
Sliders	Drag for a live change: up to ±12 h from where you start (day 0), or the full range of T. With a slider focused, each arrow key moves 1 minute.
Rows (cycles)	Automatic shows every row with data. Untick it to set the first (Min) and last (Max) row in whole cycles; negative numbers show cycles before day 0. At most 1000 rows are drawn.

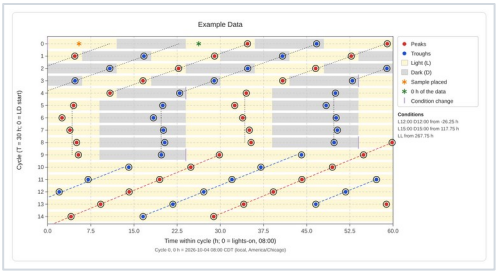
Tip. For a free-running rhythm, drag the Period T slider until the peaks line up vertically: T is then close to the free-running period (about 24:48 for the LL part of the example).

8 Reading the actogram



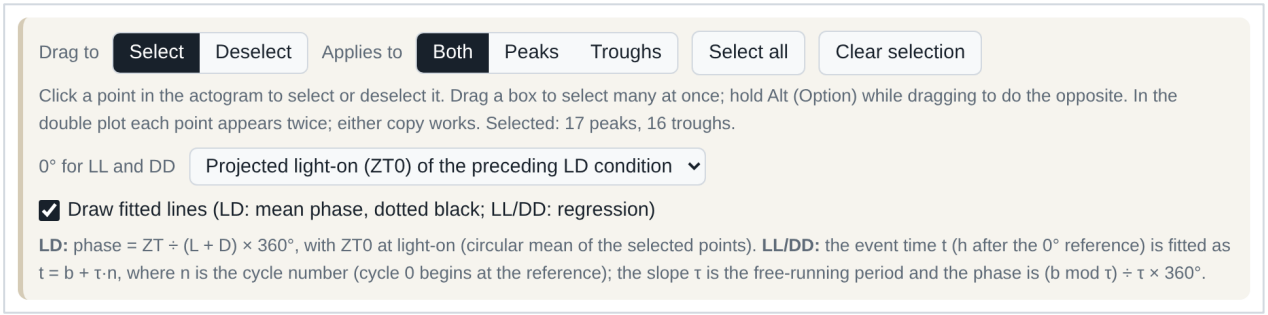
Example data with the start-time labels, the cycle start dates and 5 minor ticks

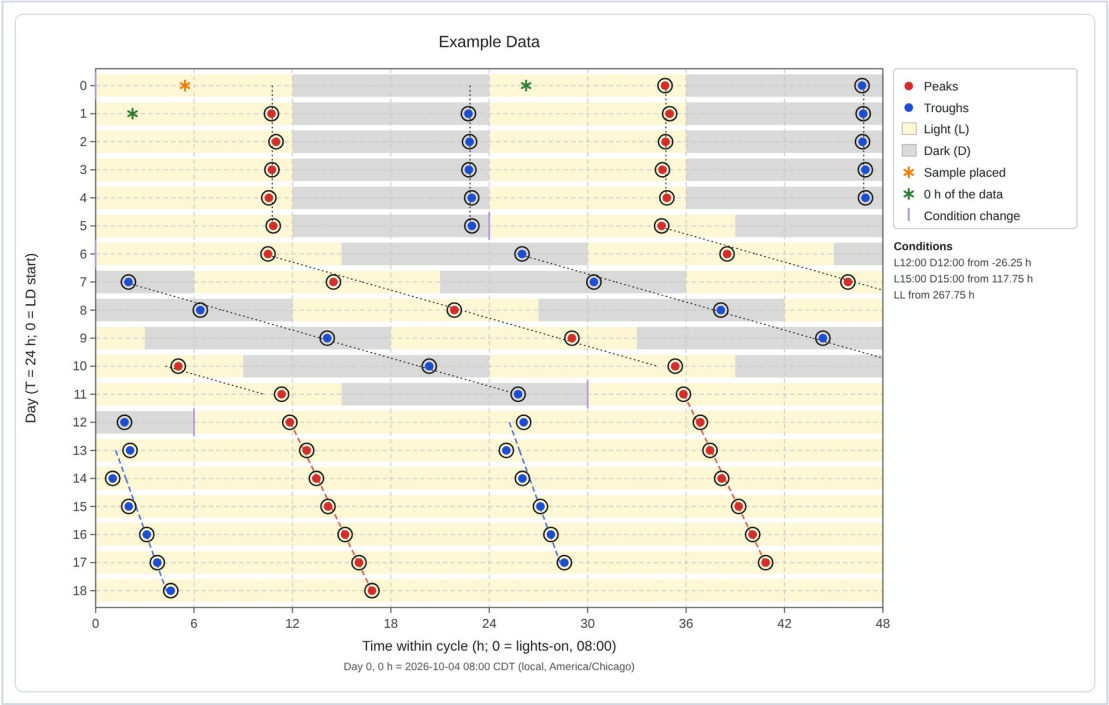
- **Rows** are successive cycles of length T, from day 0 downwards.
- **Double plot:** each row shows its own cycle and the next one, so every point appears twice (at x and, one row higher, at x + T).
- **Bands:** light and dark periods (pale yellow and grey by default; other colours under Plot).
- **Asterisks:** green = 0 h of the data, orange = sample placement. Light-purple lines: condition changes.
- **Phase analysis:** ringed points are selected; dotted black = LD mean phase, dashed = LL/DD regression.
- **Points:** red = peaks, blue = troughs. A rhythm locked to the cycle forms a vertical line; a free-running rhythm drifts to the right (period longer than T) or to the left (shorter).
- **Right side:** legend and the list of conditions with their start in hours. Under the x axis: the clock time of day 0, 0 h.



With Period T = 30:00 the LD 15:15 part of the example lines up, and the axis runs 0–60.

9 Phase analysis: selecting points

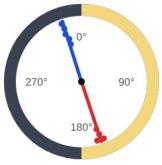




Selected points are ringed; dotted black lines: LD mean phase; dashed: LL regression

- **Click** a point in the actogram to select or deselect it. In the double plot either copy works.
- **Drag a box** to select many points; hold Alt (Option) to deselect instead, or set Drag to to Deselect.
- **Applies to** limits dragging, Select all and Clear selection to peaks, troughs or both.
- **0° for LL and DD:** projected light-on (ZT0) of the preceding LD condition, the start of the condition, 0 h of the data or day 0, 0 h.
- **Draw fitted lines** shows the LD mean phases and the LL/DD regressions in the actogram.
- In the desktop version, switch the toolbar's zoom and pan off before selecting.

9 Phase analysis: results



#1 LD 12:00/12:00

Starts 2026-10-04 08:00 CDT (-26.25 h), begins with light. ZT0 = light-on; cycle length 24.00 h.

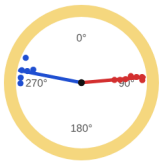
161.6° Peaks: mean ZT 10.77 h, n = 5

342.5° Troughs: mean ZT 22.83 h, n = 5

Kind	N	Mean phase (°)	Mean ZT (h) ⓘ	r ⓘ	Circular SD (°)
Peaks	5	161.6	10.77	0.999	2.2
Troughs	5	342.5	22.83	1.000	1.3

Mean ZT (h) is the phase in hours: the time after light-on (ZT0) within the 24.00-h cycle, = phase (°) ÷ 360 × (L + D).

▶ Each selected point



#3 LL

Starts 2026-10-16 14:00 CDT (267.75 h). 0° = projected light-on (ZT0) of LD #2: 2026-10-16 14:00 CDT (267.75 h).

85.1° Peaks: τ = 24.83 h (24:50), n = 7

281.1° Troughs: τ = 24.62 h (24:37), n = 6

Kind	N	Period τ (h) ⓘ	± SE	Intercept b (h) ⓘ	Phase (°)	± SE (°)	Phase (h) ⓘ	Phase (CT) ⓘ	R^2
Peaks	7	24.827	0.023	5.87	85.1	1.2	5.87	5.67	1.0000
Troughs	6	24.619	0.155	19.22	281.1	6.9	19.22	18.74	0.9998

Phase (h) is where the rhythm sits within one cycle: the hours from the 0° reference to the fitted event, b mod τ (= phase (°) ÷ 360 × τ). **Phase (CT)** is the same in circadian hours, with one cycle τ scaled to 24 h: Phase (h) × 24 ÷ τ.

▶ Each selected point

One card per condition. The dial has 0° at the top, clockwise; its ring shows light and dark for LD (all light for LL, all dark for DD); the dots are the selected points and the lines from the centre the mean phases.

Value	Meaning
Mean phase (°), Mean ZT (h)	LD: circular mean of the points. ZT = hours after light-on; phase = ZT ÷ (L + D) × 360°.
r, Circular SD (°)	LD: how tightly the points cluster (r = 1: all at one phase) and their circular spread.
Period τ (h) ± SE	LL/DD: slope of the regression t = b + τ·n of the time t (h after 0°) on the cycle number n.
Intercept b (h)	LL/DD: fitted time of cycle 0, in hours after the 0° reference.
Phase (°) ± SE, Phase (h)	LL/DD: b mod τ, as a fraction of the cycle (°) and in hours after the 0° reference.
Phase (CT)	LL/DD: Phase (h) in circadian hours, with τ scaled to 24 h: Phase (h) × 24 ÷ τ.
R^2	LL/DD: how well the straight line fits the points.

9 Phase analysis: summary and export

Condition	Type	n	Period (h)	Period SE (h)	R^2	Phase (°)	Phase (h)	Phase SD (°)	0° reference
Condition 1 · LD 12:00/12:00	Peaks	5	24.00			161.6	10.77	2.2	light-on (ZT0)
Condition 1 · LD 12:00/12:00	Troughs	5	24.00			342.5	22.83	1.3	light-on (ZT0)
Condition 2 · LD 15:00/15:00	Peaks	5	30.00			123.1	10.26	12.1	light-on (ZT0)
Condition 2 · LD 15:00/15:00	Troughs	5	30.00			308.7	25.72	8.3	light-on (ZT0)
Condition 3 · LL	Peaks	7	24.83	0.02	1.000	85.1	5.87		projected ZT0 of the preceding L
Condition 3 · LL	Troughs	6	24.62	0.15	1.000	281.1	19.22		projected ZT0 of the preceding L

- **Summary table:** one row per condition and type, below the cards. Phase (h) is the phase in hours within one cycle (LD: mean ZT; LL/DD: b mod τ).
- **Selected points:** every point with its condition, cycle, ZT or time from 0°, phase and residual.
- **Download results (CSV)** saves both tables for Excel or R; **Copy results as tab-separated text** puts them on the clipboard for pasting into a spreadsheet.
- **Settings files** also keep the selection, so a saved analysis can be reopened.
- **Desktop version:** the same tables are under the actogram; Phase results… opens the cards in a window that can be saved as an image.

How the numbers are calculated

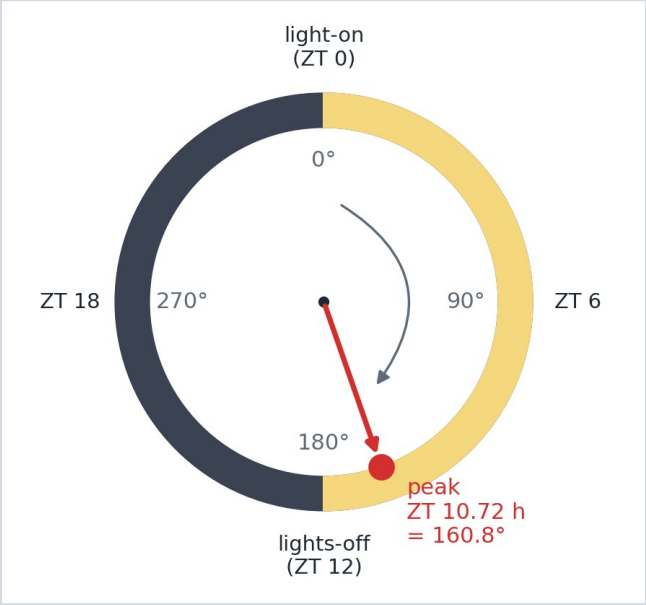
LD. ZT = time after light-on within the L + D cycle; phase = ZT ÷ (L + D) × 360°. The mean phase is the circular mean of the selected points, r the mean resultant length and the circular SD = √(−2 ln r).

LL and DD. Each point gets a cycle number n (cycle 0 begins at the 0° reference; a gap of about one period adds one cycle) and its time t in hours after the reference. A straight line $t = b + \tau \cdot n$ is fitted by least squares: τ is the free-running period, b the intercept, phase = (b mod τ) ÷ τ × 360° and Phase (CT) = (b mod τ) × 24 ÷ τ. The ± SE of the phase comes from the standard error of b.

Tip. Select only clear, well-defined peaks or troughs, and check the R^2 and SE of each LL/DD fit before reporting a period.

What is a phase?

Your samples have a rhythm: it goes up and down, again and again, like the hands of a clock going round and round. A **phase** simply tells you **where in the round** something happens, for example where the peak is.



- Think of one whole cycle as a **clock face**. In LD the top of the clock is the moment the lights go on (ZT 0).
- Going once all the way round is **one cycle**: 24 h in LD 12:12, 30 h in LD 15:15.
- **Degrees** just say how far round: a quarter = 90°, half = 180°, three quarters = 270°, all the way = 360° (back at the top).
- So hours turn into degrees like this:

degrees = hours ÷ cycle length × 360

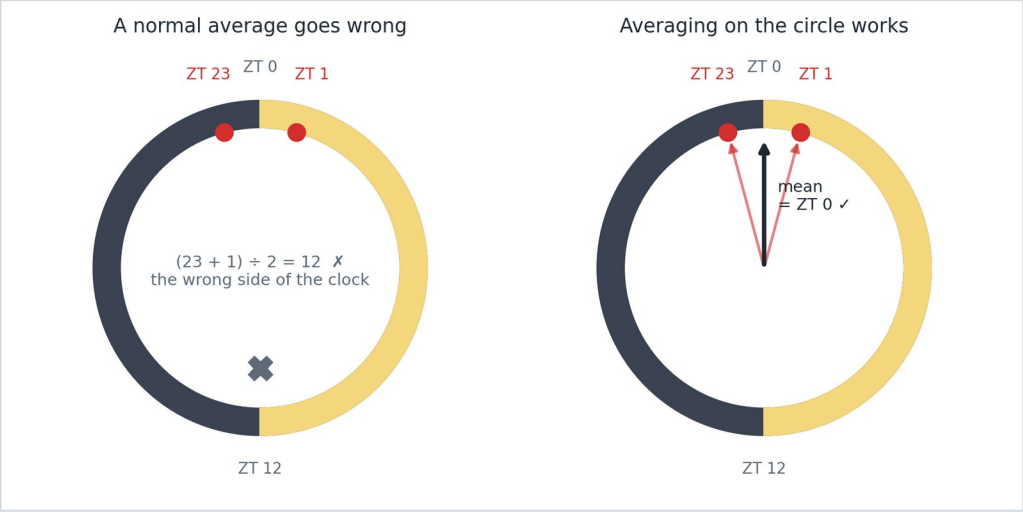
Example. In the example data the first peak comes 10.72 hours after the lights went on. 10.72 ÷ 24 × 360 = 160.8°: a little less than half way round the clock, so the peak is near the end of the light period.

Two words you will meet

- **ZT (zeitgeber time):** hours after the lights went on, in an LD condition. ZT 0 = light-on.
- **CT (circadian time):** the same idea in LL or DD, where there is no light-on to follow. One of the rhythm's own cycles is counted as 24 "circadian hours", however long it really is.
- **mod:** "take away whole cycles and keep what is left". 30 h mod 24 h = 6 h, just like 30 o'clock is 6 o'clock the next day.

Φ LD: the average phase

Why can't we just add up and divide? Because the clock goes round. A peak at ZT 23 (just before light-on) and one at ZT 1 (just after) are really close: only 2 hours apart, across the top. A normal average says ZT 12, the opposite side of the clock! So the app averages **arrows** instead of numbers.



How the app does it, step by step

1. **Find each point's ZT.**
   ZT = (time of the point − time of light-on), then take away whole cycles (mod L + D). Example: a peak 34.72 h after a light-on in LD 12:12 has ZT 34.72 mod 24 = 10.72 h.

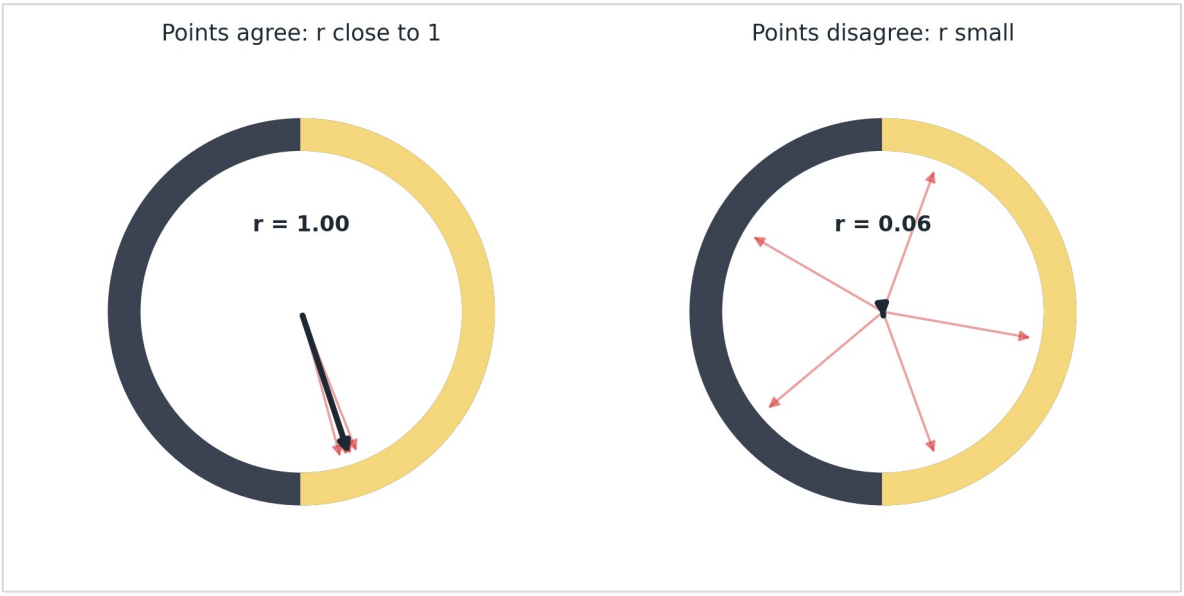
2. **Turn the ZT into an angle.**
   angle = ZT ÷ (L + D) × 360°.

3. **Draw an arrow for each point.**
   Every arrow starts at the centre of the clock, has length 1 and points at the angle of its point.

4. **Add the arrows and divide by how many there are.**
   Put them end to end (add their x parts and their y parts) and divide by N. This gives one "average arrow".

5. **Read the answer.**
   The direction of the average arrow is the Mean phase (°). Its length is r (next page). Mean ZT (h) = Mean phase ÷ 360 × (L + D).

LD: do the points agree?



- **r (mean resultant length)** is the length of the average arrow, from 0 to 1. When all the arrows point the same way they add up to a long arrow: r is close to 1 (the points agree). When they point every which way they cancel out: r is close to 0.
- **Circular SD (°)** turns r into a spread in degrees: SD = √(−2 ln r), in degrees. Small = the points are tightly together; it behaves like an ordinary standard deviation when the points are close.

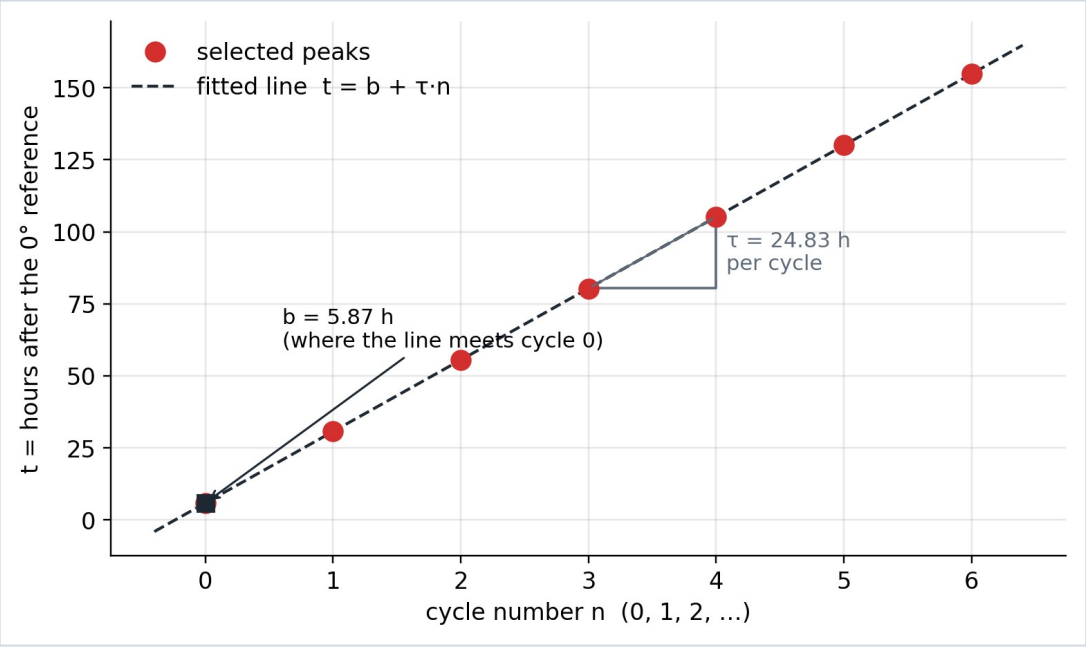
Worked example: the peaks of condition #1 (LD 12:12, light-on 08:00)

Peak	Time (h after 0 h)	Time after light-on (h)	ZT (h) = … mod 24	Angle (°) = ZT ÷ 24 × 360
1	8.47	34.72	10.72	160.8
2	32.75	59.00	11.00	165.0
3	56.50	82.75	10.75	161.3
4	80.31	106.56	10.56	158.4
5	104.58	130.83	10.83	162.4

Result. Average of the five arrows: Mean phase = 161.6°, so Mean ZT = 161.6 ÷ 360 × 24 = 10.77 h. r = 0.999 (almost 1: the five peaks agree very well) and Circular SD = 2.2°.

LL and DD: a straight line

In constant light (LL) or constant darkness (DD) there is no light-on to measure from, and the rhythm's own clock runs a little fast or slow, so every cycle the peak comes a bit earlier or later. Write down, for each cycle, how many hours after a starting point (the **0° reference**) the peak happened. The numbers go up by about the same amount each time, like stairs that all have the same height. A straight line through them tells us two things:



- **the slope τ (tau)** = the height of each stair = how many hours one cycle takes: the free-running **period**.
- **the intercept b** = where the line is at cycle 0 = when the peak "would be" in the first cycle after the reference.

How the app does it, step by step

1. **Choose the 0° reference R.** e.g. the projected light-on (ZT0) of the LD before. t = time − R for every point.

2. **Number the cycles n.** The first point gets n = floor((t of the first point) ÷ the previous LD period); each next point gets one more, or more if a cycle was skipped.

3. **Fit the straight line t = b + τ·n.** by least squares: of all straight lines, the one whose squared up-and-down distances to the points add up to the smallest number.

LL and DD: from the line to the phase

Worked example: the peaks of condition #3 (LL), 0° = 16 Oct 14:00 (267.75 h)

Cycle n	Time (h)	t = time − 267.75	On the line: b + τ·n	Residual (h)
0	273.59	5.84	5.87	-0.03
1	298.62	30.87	30.70	+0.17
2	323.21	55.46	55.52	-0.06
3	347.92	80.17	80.35	-0.18
4	372.96	105.21	105.18	+0.03
5	397.81	130.06	130.01	+0.05
6	422.60	154.85	154.83	+0.02

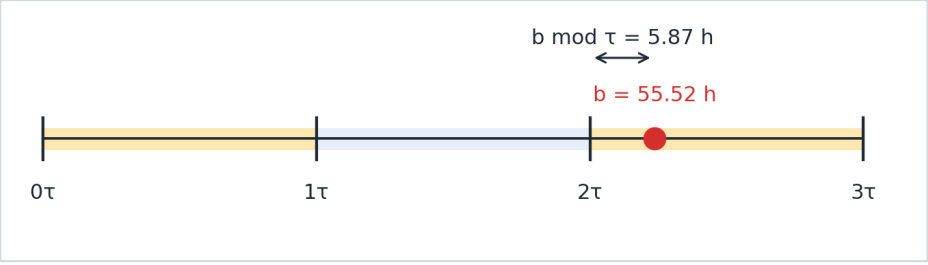
Period τ: 24.827 h: each cycle takes about 24.83 hours (0.83 h longer than a day).

Intercept b: 5.87 h: on the line, the peak of cycle 0 comes 5.87 h after the 0° reference.

Phase (h) = b mod τ: 5.87 mod 24.83 = 5.87 h. If b were bigger than one cycle (for example 55.52 h), whole cycles would be taken away first, as in the ruler below.

Phase (°) = Phase (h) ÷ τ × 360: 5.87 ÷ 24.83 × 360 = 85.1°: about a quarter of the way round.

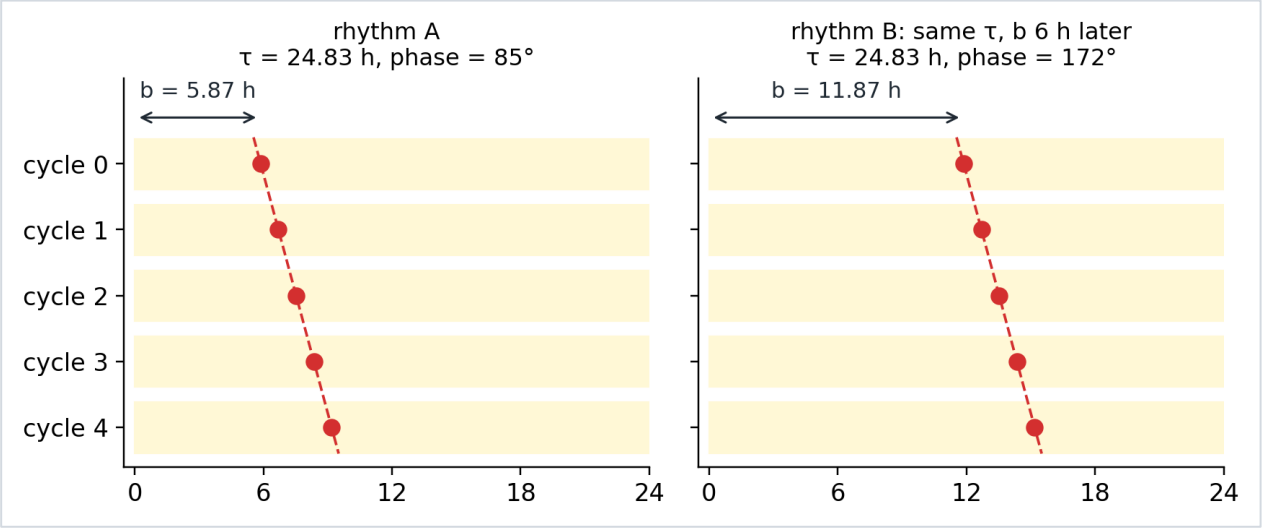
Phase (CT) = Phase (h) × 24 ÷ τ: 5.87 × 24 ÷ 24.83 = 5.67 circadian hours: the same place in the cycle, as if the cycle were stretched or squeezed to exactly 24 h.



- **± SE (standard error):** how sure we are. τ = 24.83 ± 0.023 h; the phase ± SE comes from the SE of b (0.082 h): 0.082 ÷ 24.83 × 360 = 1.2°. Repeating the experiment would usually give a value within about ± 2 SE.
- **R^2:** how straight the points are, from 0 to 1. 1 = all exactly on the line. Here R^2 = 1.0000.
- **Residual:** how far a point is above (+) or below (−) the line, in hours.

LL and DD: why the intercept b matters

The period τ only says how **fast** the rhythm's clock runs. It does not say **what time** that clock shows. Two rhythms can have exactly the same τ and still have their peaks at very different times of the cycle. The intercept b is the number that pins down **where** the rhythm is, so it is what the phase is made from.

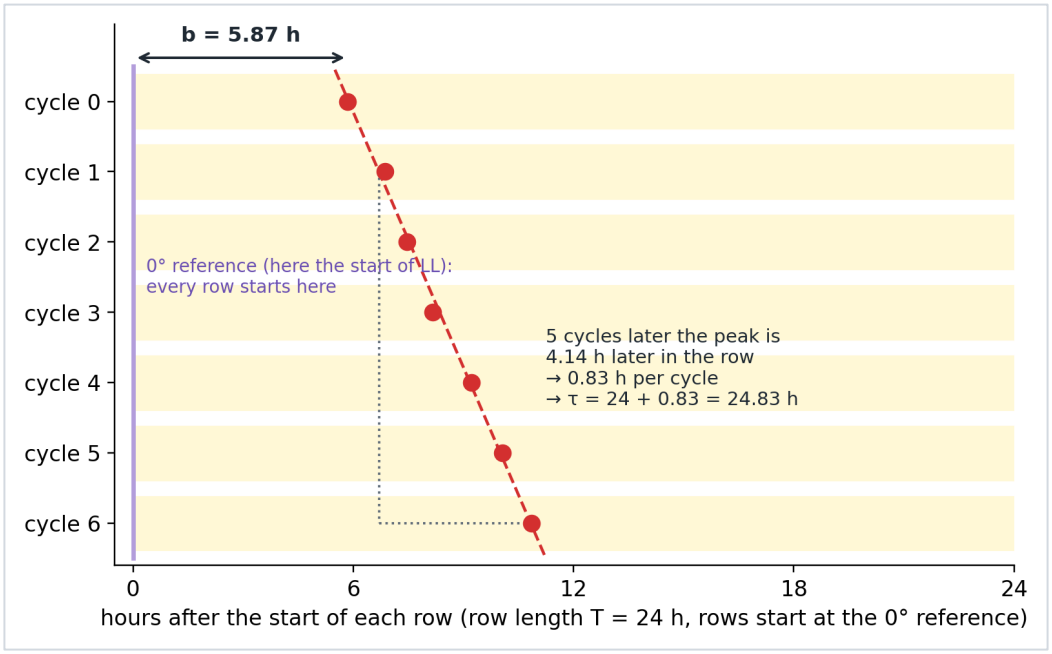


Each row is one cycle, starting at the 0° reference. Same slope (same τ), but B's line starts 6 h later: its phase is different.

- **b is measured from the 0° reference.** b = 5.87 h means: following the line back to cycle 0, the peak comes 5.87 h after the reference. The reference plus b is the clock time of that peak. Change the reference and b (and the phase) change with it, so always report which 0° you used.
- **b uses all the points, not just the first one.** A single peak can be early or late by chance (the residuals here go up to ±0.18 h). The line averages every selected cycle, so b is more reliable: its standard error is only 0.08 h.
- **b and τ are found together.** If τ were wrong, the line would tilt and b would move. That is why the line is fitted to all the points at once, and why a fit needs points in at least two cycles.
- **Whole cycles do not matter: b mod τ.** If you start counting at a later cycle, b grows by whole periods τ; taking whole cycles away (mod) gives the same phase, Phase (h) = b mod τ, whichever cycle you call 0.
- **Phase in three units.** From b mod τ: Phase (h) = 5.87 h after 0°; Phase (°) = 5.87 ÷ 24.83 × 360 = 85.1°; Phase (CT) = 5.87 × 24 ÷ 24.83 = 5.67.

Φ Period and phase on the actogram

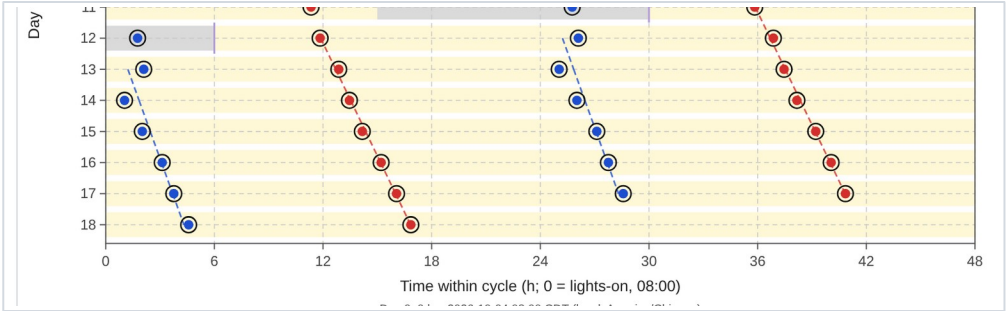
The same fit, seen as an actogram. Here each row is one 24-h window that starts at the 0° reference, so the **first row is cycle 0**. Each peak sits at t − 24·n in its row, so the fitted line becomes a straight line through the rows.



1. **Where the line crosses the first row is b.** The line meets cycle 0 at 5.87 h after the reference: that is the intercept, and (b mod τ) the phase.

2. **How much the line leans is the period.** Each row the peak moves 0.83 h to the right, so one cycle is 24 + 0.83 = 24.83 h. Leaning right: τ longer than the row (24 h); leaning left: shorter; straight down: τ = 24 h.

3. **The dots around the line are the residuals.** The closer they are to the line, the higher R^2 and the smaller the ± SE.



In the app (Draw fitted lines): the same fits as dashed lines on the double plot. Rows start at day 0, not at the 0° reference, so the line sits elsewhere in the row, but its lean (τ) and the phase are the same.

Every result, explained (1): the cards

Item	What it means	In simple words
Kind	Peaks or troughs of the condition.	Which points the row is about.
N	Number of selected points.	How many points were used.
Big red / blue number	Phase (°) of the peaks / troughs.	Where round the clock they are.
Mean phase (°) (LD)	Circular mean of the points' angles; 0° = light-on.	Direction of the average arrow.
Mean ZT (h) (LD)	Mean phase ÷ 360 × (L + D).	Hours after the lights go on.
r (LD)	Mean resultant length, 0–1.	How much the points agree (1 = perfectly).
Circular SD (°) (LD)	√(−2 ln r), in degrees.	How spread out the points are.
Period τ (h) (LL/DD)	Slope of t = b + τ·n.	How long one cycle of the rhythm takes.
± SE (after τ)	Standard error of the slope.	How sure we are about τ.
Intercept b (h) (LL/DD)	Value of the line at cycle 0, h after the 0° reference.	Where the line starts.
Phase (°) (LL/DD)	(b mod τ) ÷ τ × 360.	Where round the cycle, from 0°.
± SE (°)	SE of b ÷ τ × 360.	How sure we are about the phase.
Phase (h) (LL/DD)	b mod τ.	The phase in hours after 0°.
Phase (CT) (LL/DD)	Phase (h) × 24 ÷ τ.	The phase on a 24-hour "rhythm clock".
R^2 (LL/DD)	1 − (squared distances to the line) ÷ (squared distances to the mean).	How straight the points are.
0° = …	The 0° reference of LL/DD (see the Phase analysis settings).	Where the counting starts.
Dial	Ring: light (yellow) and dark; dots: points; line: mean phase (LD length = r).	A picture of all of the above.

Every result, explained (2): the tables

Summary table

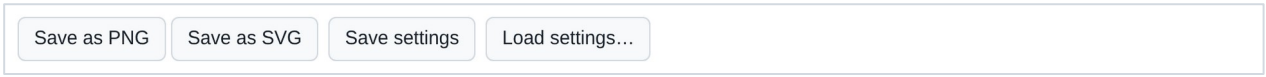
Column	What it means
Condition, Type, n	The condition (#, LD/LL/DD), peaks or troughs, and the number of points.
Period (h)	LD: the cycle length L + D. LL/DD: τ, the fitted period.
Period SE (h), R^2	LL/DD only: standard error of τ and how straight the points are.
Phase (°), Phase (h)	The phase as an angle and in hours within one cycle (LD: mean ZT; LL/DD: b mod τ).
Phase SD (°)	LD only: circular SD of the points.
0° reference	LD: light-on (ZT0). LL/DD: the chosen reference.

Selected points (“Each selected point”)

Column	What it means
Type, Time (h)	Peak or trough, and its time in hours after 0 h of the data.
Local date/time	The same moment on the wall clock (time zone of the data).
Condition	The condition the point falls in.
Cycle	LD: which cycle after the condition’s first light-on. LL/DD: the cycle number n used in the fit.
ZT or t from 0° (h)	LD: ZT. LL/DD: t, hours after the 0° reference.
Phase (°)	The point’s own phase: LD from its ZT; LL/DD from t mod τ.
Residual (h)	LL/DD: how far the point is above (+) or below (−) the fitted line.

Remember. All results use only the points you selected. If a number looks odd, check the selection first: a missing or extra point can move a mean or tilt a line.

10 Saving your work



PNG Save as PNG

A picture at three times the screen resolution, for slides, documents and posters. The file is named after the title.

SVG Save as SVG

A vector drawing that can be edited in Illustrator, Inkscape or PowerPoint (ungroup it after inserting).

{ } Save settings

All inputs and options in one JSON file: data, clock time, time zone, LD cycles and plot settings.

↑ Load settings…

Reopens a settings file. Files from the desktop version (Actogram_Plotter_app.py) open too, and vice versa.

What the browser remembers. The page keeps your display options in this browser between visits. Each visit starts again with the example data, period T = 24:00 and US Central Time, so use **Save settings** to keep your own data.

Privacy. Everything you enter stays in your browser. No data is uploaded or sent anywhere.

Time-series data. The settings file also keeps the time series, the detection settings and your hand edits, so a detection can be reopened exactly as you left it. The time-series plot has its own **Save plot as PNG** / **SVG** buttons.

Troubleshooting

Message or problem	Cause and what to do
“… does not exist in America/Chicago (the clocks jump forward then)”	The time falls in the hour skipped when daylight time starts (March). Pick another time.
“… happens twice … Choose the first or the second one”	The time falls in the hour repeated when daylight time ends (November). Choose CDT (first) or CST (second).
“Clock time of 0 h: pick a date and time.”	The clock time of 0 h is empty (for example after Clear all). Pick it with Pick….
“Select peaks or troughs in the actogram …”	No points are selected for Phase analysis. Click points or drag a box over them.
“needs at least 2 points” / “all points in one cycle”	An LL/DD fit needs points in at least two cycles. Select more points of that condition.
No asterisks	Tick “Mark 0 h of the data … and the sample placement” under Plot; the orange one also needs a sample placement time.
“Not a number: …” under a list	The list contains text, units or a decimal comma. Keep only numbers, with a decimal point.
“L and D must be between 00:00 and 99:59 …”	An LD cycle has an invalid length, or L + D = 0. Correct L or D.
“… its start must be later than the start of the previous cycle”	The changes are not in time order. Correct the start dates and times.
“Period T must be between 0:01 and 240:00”	Enter T as hh:mm within that range.
No light/dark bands	Tick “Show the LD cycles”, and fix any red message under the LD table.
Note: “The LD cycles are not valid yet, so 0 h of the data is used”	Day 0 is set to lights-on but the LD cycles have an error. Fix them, or choose another day 0 option.
Some points are missing	They lie outside the rows shown. Tick Rows → Automatic, or widen Min / Max (negative values are allowed).
“Only the first 1000 rows are drawn”	Period T is very short compared with the data. Use a longer T.
“Saving isn’t available in this view”	Open the app from its web address (dxda6216.github.io/actogram_plotter) and save from there.

Troubleshooting: time-series data

Message or problem	Cause and what to do
“No time-series data yet.”	The data box is empty. Paste the data, use Open file..., or Load example series.
“Only n data rows were read; at least 10 are needed.”	Fewer than 10 rows with two numbers. Check the separators and use a decimal point, not a comma.
“n lines skipped (not two numbers)”	Header or note lines were skipped, which is fine. If data rows are skipped, remove units or text from them.
“The cutoff period must be longer than 2 sampling intervals” / “... must not exceed 240 h”	Choose a cutoff period between two intervals and 240 h (48 h is usual).
“The sinc filter order must be a whole number of at least 3.”	Enter 3 or more (odd numbers; 101 is the default).
“The moving-average window must be at least 3 sampling intervals” / “... is longer than the recording”	Change the Trend window (24 h is usual).
“Detrending failed: ... too short for a filter of n taps”	The record is too short for the filter order. Lower the order, use Moving Average, or detect in the raw data.
Extra peaks or troughs from small wiggles	Use a longer moving average (Line) or a larger Min. peak spacing, or remove them by hand.
Hours start at the wrong place (date/time data)	Check the Clock time of 0 h, the UTC offsets in column 1 and the Time zone (rows without an offset use it).
A clear peak is missing	Its neighbour may be within the minimum spacing; lower the spacing, or add the peak by hand. The first and last point cannot be a peak.
The times differ slightly from another program	Unequal intervals are resampled onto a grid and the line is a moving average; check the settings on page 10.
“No marker near the click.”	Click closer to the marker, or zoom in first.
“... is already marked as the opposite type; remove it first.”	That sample is already a trough (or peak). Remove it, then add the other kind.
No L/D bars over the time-series plot	Tick Show the light/dark conditions and fix any red message under the LD cards and the clock time of 0 h.
The peak lists cannot be edited	In Detect in time-series data the lists are read-only: edit in the plot, or switch to Enter the times.